Endomorphism rings & modules

(V, T) : k-pair

$\Updownarrow$

"A vector space with scalar multiplication by x"

$\Updownarrow$

"An abelian group with scalar multiplication by k & by x"

$\Updownarrow$

"An abelian group with scalar multiplication by $k[x]$"

Flashback : $G \curvearrowright X$

$G \times X \longrightarrow X$ $(g, x) \longmapsto g \cdot x$ $\vdots$	$\rho : G \longrightarrow \underbrace{\mathrm{Bij}(X)}$ group homomorphism

Def^{n}: M : additive abelian group

An **endomorphism of M** is a group homomorphism $f : M \longrightarrow M$

$$\mathrm{End}(M) = \{ f : M \longrightarrow M \}$$

$$\in$$

$$0_M : m \longmapsto 0, \ \forall m \in M$$

Observation:

$\mathrm{End}(M)$ has **two operations**:

(a) (Addition) $f_1, f_2 \in \mathrm{End}(M)$, $m \in M$

$$0_M \xleftarrow{\text{identity}} \qquad (f_1 + f_2)(m) = f_1(m) + f_2(m)$$

(b) (Composition) $f_1, f_2 \in \mathrm{End}(M)$

$$id_M \xleftarrow{\text{identity}} \qquad (f_1 \circ f_2)(m) = f_1(f_2(m))$$

$$[\ (f_1 \circ f_2)(m_1 + m_2) = f_1(f_2(m_1 + m_2)) = f_1(f_2(m_1)) + f_1(f_2(m_2))\ ]$$

Properties:

(a) Addition is commutative

(b) End(M) is an abelian group under addition

Inverses are given by

$$f \longmapsto (-f : m \mapsto -f(m))$$

Associativity is guarenteed by associativity of addition in M

(c) Composition is associative

(d) Composition distributes over addition

$$f_1 \circ (f_2 + f_3) : m \mapsto f_1((f_2+f_3)(m)) = f_1(f_2(m)) + f_1(f_3(m)) = ((f_1 \circ f_2) + (f_1 \circ f_3))(m)$$

Composition is not necessarily commutative.

Upshot: $\mathrm{End}(M)$ is a (possibly non-commutative) ring

Examples:

(1) $\mathrm{End}(\mathbb{Z}) \simeq \mathbb{Z}$

$$\Phi : f \longmapsto f(1)$$

Pf: (a) $\Phi(id_{\mathbb{Z}}) = id_{\mathbb{Z}}(1) = 1$ ✓

(b) $\Phi(f_1 + f_2) = (f_1 + f_2)(1)$

$= f_1(1) + f_2(1)$

$= \Phi(f_1) + \Phi(f_2)$

(c) $\Phi(f_1 \circ f_2) = f_1(f_2(1))$

$= f_1(f_2(1) \cdot 1)$

$$\boxed{= f_2(1) f_1(1)}$$

$$= f_1(1) f_2(1)$$

$$= \Phi(f_1)\Phi(f_2)$$

(2) $\mathrm{End}(\mathbb{Z}^n) = \{ f : \mathbb{Z}^n \to \mathbb{Z}^n \}$

$$\simeq \underbrace{M_{n\times n}(\mathbb{Z})}_{\text{ring via matrix addition \& multiplication}}$$

$f \mapsto [f(e_1) \cdots f(e_n)] \in M_n(\mathbb{Z})$

Non-commutative if $n \geq 1$.

Modules

Def$^{\underline{n}}$ R: ring

A module over R is a pair (M, φ) where:

- M is an additive abelian group
- $\varphi: R \to \text{End}(M)$ is a ring homomorphism

<u>Fact:</u> M is an additive abelian group.

Then extending M to an R-module (M, φ)

$\Leftrightarrow$ Giving a function

$$R \times M \longrightarrow M$$
$$(r, m) \longmapsto r \cdot m \quad (= \varphi(r)(m))$$

<u>s.t.</u>

(a) $r \cdot (m_1 + m_2) = r \cdot m_1 + r \cdot m_2$

$(\Leftrightarrow \varphi(r) \in \text{End}(M))$

(b) $1 \cdot m = m$

$(\Leftarrow)\ \varphi(1) = id_M$)

(c) $(r_1 + r_2) \cdot m = r_1 \cdot m + r_2 \cdot m$

$(\Leftarrow)\ \varphi(r_1 + r_2) = \varphi(r_1) + \varphi(r_2)$)

(d) $r_1 \cdot (r_2 \cdot m) = (r_1 r_2) \cdot m$

$(\Leftarrow)\ \varphi(r_1) \circ \varphi(r_2) = \varphi(r_1 r_2)$)

Proposition

$$\left\{ \begin{array}{c} k\text{-pairs} \\ (V, T) \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{c} \text{Modules} \\ \text{over} \\ k[x] \end{array} \right\}$$